

7. SIGNAL RECONSTRUCTION AND SPECTRAL ANALYSIS

Objective

This experiment is divided into four simulations:

- 1. Reconstruction of Signal from Samples (Ideal Interpolation)**
- 2. Fourier Series of Half-Wave Rectifier Output**
- 3. Fourier Transform of a Continuous-Time Signal**
- 4. Fourier Transform of a Rectangular Waveform**

Experiment 7.1

Reconstruction of Signal from Samples (Ideal Interpolation)

Aim

Reconstruct a bandlimited signal using sinc interpolation.

Program (MATLAB)

```
% Reconstruction of Signal from Samples
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Original signal frequency
```

```
fm = 5; % Hz
```

```
% Sampling frequency
```

```
fs = 25; % Hz
```

```
% Continuous-time signal
```

```
t = 0:0.001:1;
```

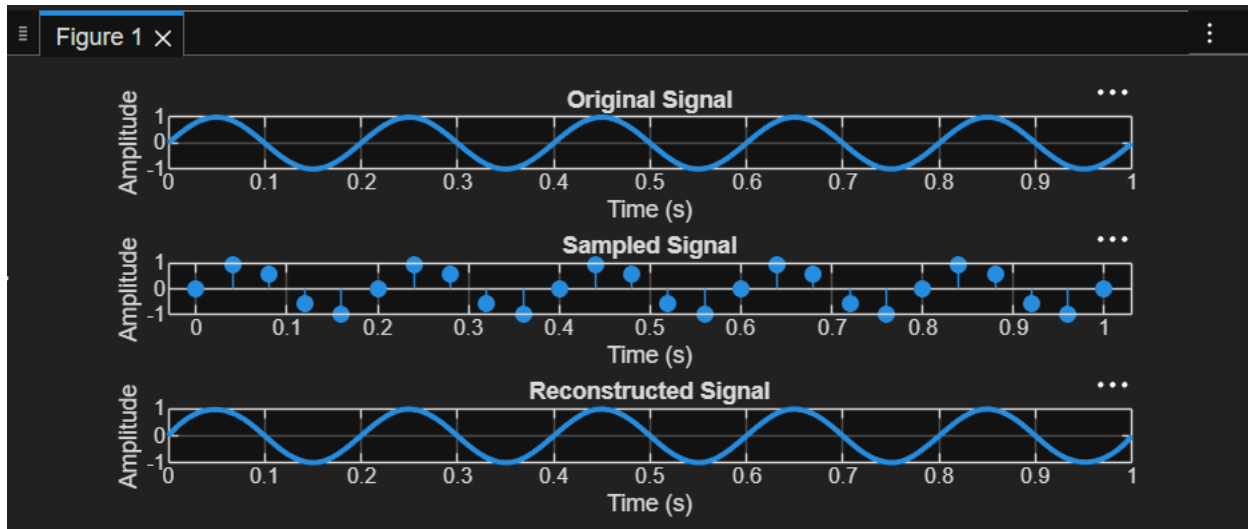
```
x = sin(2*pi*fm*t);
```

```
% Sampling
ts = 0:1/fs:1;
xs = sin(2*pi*fm*ts);
% Reconstruction using interpolation
xr = interp1(ts,xs,t,'spline');
% Plotting
figure;
subplot(3,1,1);
plot(t,x,'LineWidth',2);
grid on;
title('Original Signal');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,2);
stem(ts,xs,'filled');
grid on;
title('Sampled Signal');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,3);
plot(t,xr,'LineWidth',2);
grid on;
title('Reconstructed Signal');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

Output Generated Graph:



Result

The sinusoidal signal was sampled at 25 Hz, which is greater than the Nyquist rate of 10 Hz. Using interpolation, the original signal was reconstructed successfully, demonstrating accurate signal recovery from its samples in accordance with the Sampling Theorem.

Experiment 7.2

Fourier Series of Half-Wave Rectifier Output

Aim:

To compute and analyze the Fourier series representation of the output of a half-wave rectifier. The Aim is to understand how the Fourier series can decompose a non-sinusoidal periodic signal into its constituent sinusoidal components (harmonics), and to observe how these harmonics contribute to the overall signal.

Program:

```
% Fourier Series of Half-Wave Rectifier Output

clc;

clear;

close all;

% Time axis
t = 0:0.001:2*pi;

% Half-wave rectified signal
x = sin(t);
x(x < 0) = 0;

% Fourier Series approximation
N = 10; % Number of harmonics
a0 = (1/pi)*trapz(t,x);
fs_signal = a0/2;

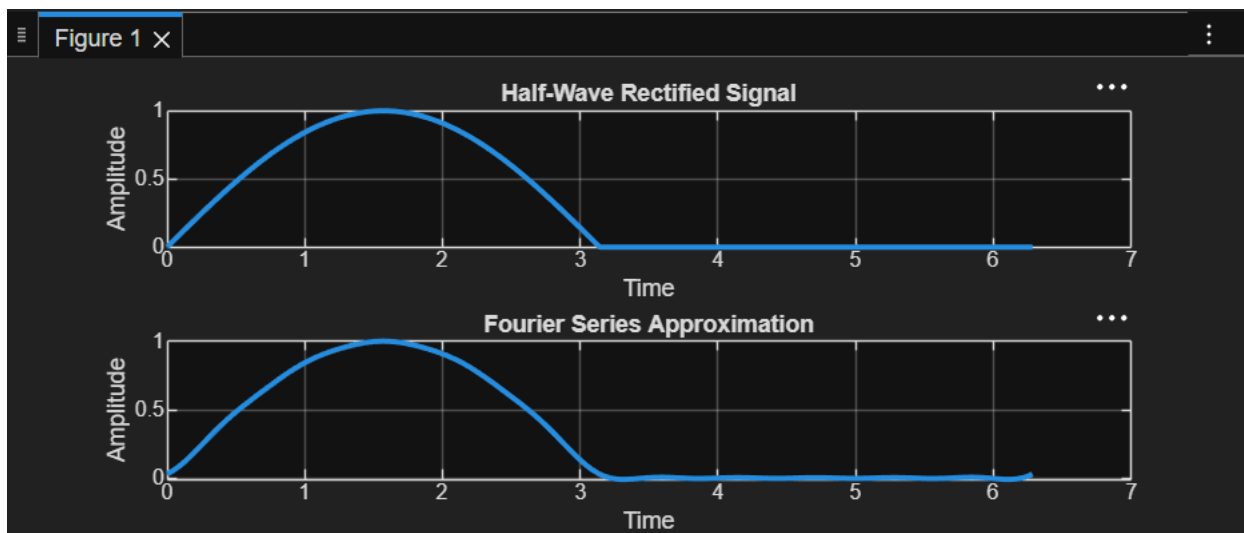
for n = 1:N
    an = (1/pi)*trapz(t,x.*cos(n*t));
    bn = (1/pi)*trapz(t,x.*sin(n*t));

    fs_signal = fs_signal + an*cos(n*t) + bn*sin(n*t);
end

% Plotting
figure;
subplot(2,1,1);
plot(t,x,'LineWidth',2);
```

```
grid on;  
title('Half-Wave Rectified Signal');  
xlabel('Time');  
ylabel('Amplitude');  
subplot(2,1,2);  
plot(t,fs_signal,'LineWidth',2);  
grid on;  
title('Fourier Series Approximation');  
xlabel('Time');  
ylabel('Amplitude');
```

Output Waveform:



Result:

The Fourier Series of the half-wave rectified signal was obtained using MATLAB. The reconstructed waveform using Fourier coefficients closely matched the original half-wave rectified signal, demonstrating the Fourier Series representation of periodic signals.